

# **Stat 414 Notes**

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# Course Goals for STAT 414

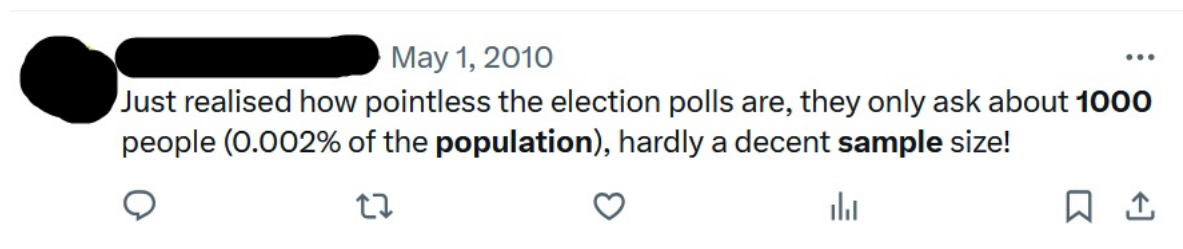
STAT 414 is an introduction to sampling. By the end of the course, you will:

- Learn the basic methodology of how classical survey samples are conducted based on simple random sampling.
- Recognize sampling and non-sampling errors.
- Be able to use extra information in sampling via ratio estimators and regression estimators.
- Be able to define and use stratified samples and know why they are often better than SRS.
- Be able to design one and two stage cluster samples and know their key properties.
- Recognize strengths and weaknesses of different sampling approaches.
- Use standard R packages to analyze survey data and estimate population means, proportions, totals, and more.

# 1 Introduction to Sampling

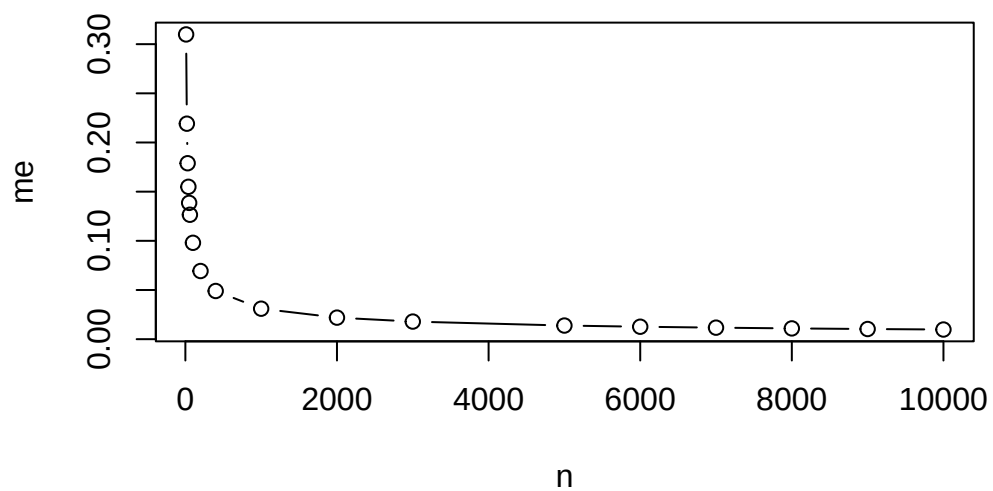
## 1.1 Introduction

If you spend any time at all on social media (especially in an election year), you're likely to come across a post like this:



What do you think? Do you agree?

```
n<-c(10,20,30,40,50,60,100,200,400,1000,2000,3000,5000,6000,7000,8000,9000,10000)
me<-1.96*sqrt(0.25/n)
plot(n,me,type="b")
```



So why even sample?

1. Cheaper
2. Faster
3. **More Accurate** (?!?!?)

  Aug 26, 2020 ...

Wednesday open question: What's one thing that's very hard for most people to believe, but which is actually true?

 414    101    241         

  ...

a surveyed sample of 1,000 individuals, if representative of the larger population, is sufficient to make accurate predictions about a population of 300 million people

3:18 PM · Aug 26, 2020

What's key here?

What we will spend this semester studying is how to ensure this, to make sure we can make accurate predictions. This will involve answering questions focused on:

- Survey design
- Preparation
- Collect and prepare data
- Data analysis

How is sampling different from previous courses, like STAT 212?

How is sampling different from previous courses, like STAT 262 or STAT 380?

How is sampling different from previous courses, like STAT 102 or STAT 318?

However, like previous courses, sample surveys do have an overarching process, some of which overlap with steps/processes we've seen before. We'll go through the principal considerations in a sample survey, along with some definitions and (possibly) new notation.

## 1.2 Considerations in planning and executing a survey

1. Objectives of the survey
2. **Defining the population** In sampling, the word **population** needs some clarification. Generally, it means the larger set from which the sample is chosen, and it's the larger group we wish to make inference/draw conclusions about. And a **sample** is a subset of the population. However, we must distinguish between
  - **Target population**
  
  - **Sampled population**

Ideally, these two overlap completely, but this does not always occur. Plus, we also need to consider the **sampling frame population**

I love the figure on page 5 of the book:

Within the **sampling frame** we have

- **Observation unit**
- **Sampling unit**

For example,

3. **Identifying potential sources of bias and ameliorating them:** In previous classes, like STAT 262 or STAT 380, we use the word **bias** to mean an estimator that systematically over- or under-estimates the true population characteristic. In sample surveys, we still use the same definition, but now biased estimators typically arise for two reasons:

- Population characteristic of interest (Estimation bias):
- Selection and/or measurement bias: Selection bias occurs when there is a mismatch between the sampled population and the target population. Measurement bias occurs when the survey observation tends to differ from the truth in a particular direction.

We'll talk about each of these in more detail.

## Selection bias

- Convenience samples
- Judgment samples
- Self-selected samples
- Undercoverage
- Overcoverage
- Nonresponse

## Measurement bias

Why can it occur?

- Unavoidable Reasons
- People!
- Survey mode: method used to distribute and gather responses

So we're concerned about response bias—what now? Careful question design is key. We could spend weeks on just questionnaire design, but will only do a very brief overview. There are other resources on campus that can help with this aspect of surveys (Bureau of Sociological Research).

We'll start with types of questions to avoid, and then move on to other considerations when writing questions.

**Common types of biased questions:**

- Leading questions

How to fix it:

- Assumptive questions

How to fix it:

- Double-barreled questions

How to fix it:

- Double negatives

How to fix it:

**Other considerations:**

- Pay careful attention to answer scale options

- Check formatting
- Avoid jargon, and be specific
- Open or closed questions?
- Question order matters

Probably most important, test questions ahead of time with the target population.

#### **4. Choosing the sampling design/data collection approach**

## 5. Choosing the analysis approach

## 6. Data analysis

7. **What's next?** The more information we have about a population, the easier it is to devise a sampling plan that will give accurate estimates. Any past survey can serve as a guide for future samples, as it will provide information about variability and cost.

The rest of the semester

## 2 An Introduction to Probability Samples

Before we can jump into sampling designs, we need to establish a few basic concepts and notation. The sampling designs we're going to discuss are **probability samples**.

💡 Definition: Probability sample

A **probability sample**

Throughout this semester we'll discuss three different ways to design a probability sample, as well as the properties of estimators arising from these designs. Probability samples allow us to avoid selection biases and help ensure the sample is representative of the population, as probability samples take selection choice away from the researcher, and put it in the hands of random selection. The three major designs we'll discuss are:

- Simple random sampling
- Stratified random sampling
- Cluster sampling

We'll also consider **systematic sampling**

To simplify things, we'll make the following assumptions:

Depending on how far we get, we may revisit relaxing some of these assumptions at the end of the semester. In all cases, our goal is to learn about some characteristic of the target population

and we'll use our sample data to estimate that characteristic. We'll also estimate standard errors, and use those to construct confidence intervals for the unknown population characteristic. We'll also discuss determining the necessary sample size.

## 2.1 Probability Sampling Definitions and Notation

Throughout this semester, we'll assume a finite population, or **universe** of  $N$  units:

Any sample we select from  $\mathcal{U}$  will be a subset of  $\mathcal{U}$ , and we'll denote the specific sample chosen as  $\mathcal{S}$ .  $\mathcal{S}$  contains  $n$  elements from  $\mathcal{U}$ .

**Example:** Suppose  $\mathcal{U} = \{1, 2, 3, 4, 5\}$  and we consider samples of size  $n = 3$ . There are 10 possible samples that could be selected:

Remember that in **probability sampling**

Because each possible sample  $\mathcal{S}$  has a known probability of being selected, each unit in the universe  $\mathcal{U}$  also has a known probability of being in the selected sample.

Let  $y_i$  be

We consider  $y_i$  to be a *fixed quantity*.

## 2.2 Probability Samples: Inference Basics

We want to use the data from a sample to estimate some characteristic of the population. Common population characteristics of interest are

**Example:** Earlier we considered the universe  $\mathcal{U} = \{1, 2, 3, 4, 5\}$ . Suppose the values of  $y_i$  are

|       |   |   |   |   |   |
|-------|---|---|---|---|---|
| $i$   | 1 | 2 | 3 | 4 | 5 |
| $y_i$ | 2 | 4 | 4 | 6 | 6 |

We also considered taking samples of size  $n = 3$  from  $\mathcal{U}$ . There are 10 possible samples, and all are equally likely.

Just like in other classes like STAT 102, 380, 212 our inference methods are going to rely on the **sampling distribution** of a statistic.

 Definition: Sampling Distribution

The **sampling distribution**

**Example, cont.:**

From the sampling distribution, we can find properties of the estimator we're using.

- **Expected value:**

- Estimation bias:

- Variance:

- Mean squared error:

### 3 Simple Random Samples

**Simple random sampling** (SRS) is the most basic probability sampling design, and it is also used in combination with other designs like stratified and cluster sampling.

💡 Definition: Simple Random Sample

A **simple random sample (SRS)**

This definition implies

Specifically, the **inclusion probability** for any sampling unit is

Note that the converse is not true!

This is called an **equal probability** design, and the SRS is only one example of equal probability design. Some stratified and cluster sampling designs also have an equal inclusion probability for all sampling units.

### 3.1 SRS: Selecting the Sample

There are two types of selection procedures, depending on whether or not the sampling units are returned to the population after they have been drawn:

- SRSWR
- SRSWOR

Conceptually, an SRS (SRSWOR) is selected as follows:

- 
- 
- 
- 
- ⋮
-

This is useful to see why each unit has the same inclusion probability.

But, this is not an especially realistic way to pick a sample. In practice, an SRS (SRSWOR) is selected as follows:

- Create a sampling frame
- Determine a selection procedure that performs SRSWOR
- A sampling unit is included in the sample if its index corresponds to one of the  $n$  random numbers selected.

So what are some selection procedures that perform SRSWOR?

- Lottery
- [random.org](https://www.random.org)

The screenshot shows the Random.org website interface. At the top, the logo 'RANDOM.ORG' is displayed in large, bold, black letters. To the right of the logo is a search bar with the text 'Search RANDOM.ORG' and a 'Search' button. Below the logo, the text 'True Random Number Service' is visible. The main heading is 'Random Integer Set Generator'. Below this, a paragraph explains that the form allows generating random sets of integers from atmospheric noise. The 'Step 1: The Sets' section shows input fields for 'Generate 1 set(s) with 20 unique random integer(s) in each.' and 'Each integer should have a value between 1 and 100 (both inclusive; limits ±1,000,000,000)'. The 'Step 2: Display Options' section includes three checked checkboxes: 'Number the sets sequentially', 'Use commas to separate the set members', and 'Sort the members of each set in ascending order'. It also has three radio buttons for the order of sets: 'Print the sets in the order they were generated' (selected), 'Order the sets by the values that occur in them', and 'Print the sets in random order'. The 'Step 3: Go!' section has a message 'Be patient! It may take a little while to generate your sets...' and three buttons: 'Get Sets', 'Reset Form', and 'Switch to Advanced Mode'.

**RANDOM.ORG** Search RANDOM.ORG Search  
True Random Number Service

### Random Integer Set Generator

This form allows you to generate random sets of integers. The randomness comes from atmospheric noise, which for many purposes is better than the pseudo-random number algorithms typically used in computer programs.

#### Step 1: The Sets

Generate  set(s) with  unique random integer(s) in each.

Each integer should have a value between  and  (both inclusive; limits  $\pm 1,000,000,000$ ).

The total number of integers must be no greater than 10,000.

#### Step 2: Display Options

Each set will be printed on a separate line. You can choose from the following extra options:

- ☒ Number the sets sequentially
- ☒ Use commas to separate the set members
- ☒ Sort the members of each set in ascending order

You can select the order in which the sets are printed:

- ☒ Print the sets in the order they were generated
- ☐ Order the sets by the values that occur in them (in this case, you should also consider sorting the members of each set)
- ☐ Print the sets in random order

#### Step 3: Go!

Be patient! It may take a little while to generate your sets...

Figure 3.1: Random.org settings to generate a SRSWOR

- Statistical software
  - SAS: `proc surveyselect method=srs;`
  - Base R: `sample(1:N, n, replace=FALSE)`
  - R packages: Need to install and load the package `sampling`, then `sample<-srswor(n,N)`. You can then use the `getdata` function to extract the sample from the population.

**Example:** The dataset `baseball.csv` in Canvas contains statistics on 797 baseball players from the rosters of all major league teams in November, 2004. We'll treat this as our population, and take an SRS of 150 players.

First, we'll read the data into R, print out the first few rows, and check the size of the population:

```
baseball<-read.csv("baseball.csv",header=TRUE)
head(baseball)
```

|                                                                             | team | leagueid | player   | salary   | pos | gplay | gstart | inning | putout | assist | error |
|-----------------------------------------------------------------------------|------|----------|----------|----------|-----|-------|--------|--------|--------|--------|-------|
| 1                                                                           | ANA  | AL       | anderga0 | 6200000  | CF  | 112   | 92     | 2375   | 211    | 5      | 2     |
| 2                                                                           | ANA  | AL       | colonba0 | 11000000 | P   | 3     | 34     | 625    | 8      | 30     | 3     |
| 3                                                                           | ANA  | AL       | davanje0 | 375000   | CF  | 108   | 27     | 743    | 75     | 1      | 0     |
| 4                                                                           | ANA  | AL       | donnebr0 | 375000   | P   | 5     | 0      | 126    | 2      | 2      | 0     |
| 5                                                                           | ANA  | AL       | eckstda0 | 2150000  | SS  | 142   | 136    | 3575   | 198    | 309    | 6     |
| 6                                                                           | ANA  | AL       | erstada0 | 7750000  | 1B  | 125   | 124    | 3196   | 986    | 66     | 4     |
| dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb csteal |      |          |          |          |     |       |        |        |        |        |       |
| 1                                                                           | 1    | -9       | 112      | 442      | 57  | 133   | 20     | 1      | 14     | 75     | 2     |
| 2                                                                           | 4    | -9       | 3        | 3        | 0   | 0     | 0      | 0      | 0      | 0      | 0     |
| 3                                                                           | 1    | -9       | 108      | 285      | 41  | 79    | 11     | 4      | 7      | 34     | 18    |
| 4                                                                           | 0    | -9       | 5        | 0        | 0   | 0     | 0      | 0      | 0      | 0      | 0     |
| 5                                                                           | 75   | -9       | 142      | 566      | 92  | 156   | 24     | 1      | 2      | 35     | 16    |
| 6                                                                           | 83   | -9       | 125      | 495      | 79  | 146   | 29     | 1      | 7      | 69     | 16    |
| walk strikeout intwalk hbp sacrhith sacrfly gidplay                         |      |          |          |          |     |       |        |        |        |        |       |
| 1                                                                           | 29   |          | 75       | 6        | 1   |       | 0      | 3      |        | 3      |       |

|   |    |    |   |    |    |   |    |
|---|----|----|---|----|----|---|----|
| 2 | 0  | 1  | 0 | 0  | 0  | 0 | 0  |
| 3 | 46 | 54 | 2 | 0  | 1  | 5 | 2  |
| 4 | 0  | 0  | 0 | 0  | 0  | 0 | 0  |
| 5 | 42 | 49 | 1 | 13 | 14 | 2 | 11 |
| 6 | 37 | 74 | 1 | 4  | 3  | 4 | 9  |

```
N<-nrow(baseball)
N
```

```
[1] 797
```

Now we'll load the necessary package, set a seed and draw a sample of 150 players:

```
library(sampling)

set.seed(98263476)
n<-150
index<-srswor(n,N)
index[1:10]
```

```
[1] 0 0 0 0 0 0 0 1 0 1
```

So the `srswor` function only gives us a 0/1 for each element in the population. To get the sample, there are a couple of options:

```
baseballsrs<-getdata(baseball,index)
head(baseballsrs)
```

|    | ID_unit | team  | leagueid | player    | salary  | pos | gplay | gstart  | inning    | putout  | assist |         |
|----|---------|-------|----------|-----------|---------|-----|-------|---------|-----------|---------|--------|---------|
| 8  | 8       | ANA   | AL       | figgich0  | 320000  | 3B  | 148   | 80      | 2116      | 57      | 129    |         |
| 10 | 10      | ANA   | AL       | greggke0  | 301500  | P   | 5     | 0       | 263       | 2       | 5      |         |
| 21 | 21      | ANA   | AL       | rodriifr0 | 375000  | P   | 3     | 0       | 252       | 5       | 9      |         |
| 31 | 31      | ARI   | NL       | cintral0  | 335000  | SS  | 154   | 125     | 3297      | 141     | 383    |         |
| 33 | 33      | ARI   | NL       | daiglca0  | 300000  | P   | 11    | 10      | 147       | 5       | 6      |         |
| 50 | 50      | ARI   | NL       | sexsori0  | 8725000 | 1B  | 23    | 23      | 613       | 199     | 27     |         |
|    | error   | dplay | pball    | gbat      | atbat   | run | hit   | seabase | thirdbase | homerun | rbi    | stolenb |
| 8  | 11      | 9     | -9       | 148       | 577     | 83  | 171   | 22      | 17        | 5       | 60     | 34      |
| 10 | 0       | 1     | -9       | 5         | 0       | 0   | 0     | 0       | 0         | 0       | 0      | 0       |
| 21 | 0       | 0     | -9       | 3         | 0       | 0   | 0     | 0       | 0         | 0       | 0      | 0       |
| 31 | 15      | 61    | -9       | 154       | 564     | 56  | 148   | 31      | 7         | 4       | 49     | 3       |
| 33 | 0       | 0     | -9       | 11        | 17      | 2   | 2     | 2       | 0         | 0       | 0      | 0       |
| 50 | 1       | 19    | -9       | 23        | 90      | 20  | 21    | 4       | 0         | 9       | 23     | 0       |

|    | csteal | walk | strikeout | intwalk | hbp | sacrhit | sacrfly | gidplay |
|----|--------|------|-----------|---------|-----|---------|---------|---------|
| 8  | 13     | 49   | 94        | 0       | 0   | 10      | 2       | 6       |
| 10 | 0      | 0    | 0         | 0       | 0   | 0       | 0       | 0       |
| 21 | 0      | 0    | 0         | 0       | 0   | 0       | 0       | 0       |
| 31 | 3      | 31   | 59        | 2       | 2   | 12      | 4       | 11      |
| 33 | 0      | 0    | 7         | 0       | 0   | 1       | 0       | 1       |
| 50 | 0      | 14   | 21        | 0       | 0   | 0       | 0       | 2       |

```
baseballsrs2<-baseball[(1:N)[index==1],]
head(baseballsrs2)
```

|    | team | leagueid | player    | salary  | pos | gplay | gstart | inning | putout | assist | error |
|----|------|----------|-----------|---------|-----|-------|--------|--------|--------|--------|-------|
| 8  | ANA  | AL       | figgich0  | 320000  | 3B  | 148   | 80     | 2116   | 57     | 129    | 11    |
| 10 | ANA  | AL       | greggke0  | 301500  | P   | 5     | 0      | 263    | 2      | 5      | 0     |
| 21 | ANA  | AL       | rodriifr0 | 375000  | P   | 3     | 0      | 252    | 5      | 9      | 0     |
| 31 | ARI  | NL       | cintral0  | 335000  | SS  | 154   | 125    | 3297   | 141    | 383    | 15    |
| 33 | ARI  | NL       | daiglca0  | 300000  | P   | 11    | 10     | 147    | 5      | 6      | 0     |
| 50 | ARI  | NL       | sexsori0  | 8725000 | 1B  | 23    | 23     | 613    | 199    | 27     | 1     |

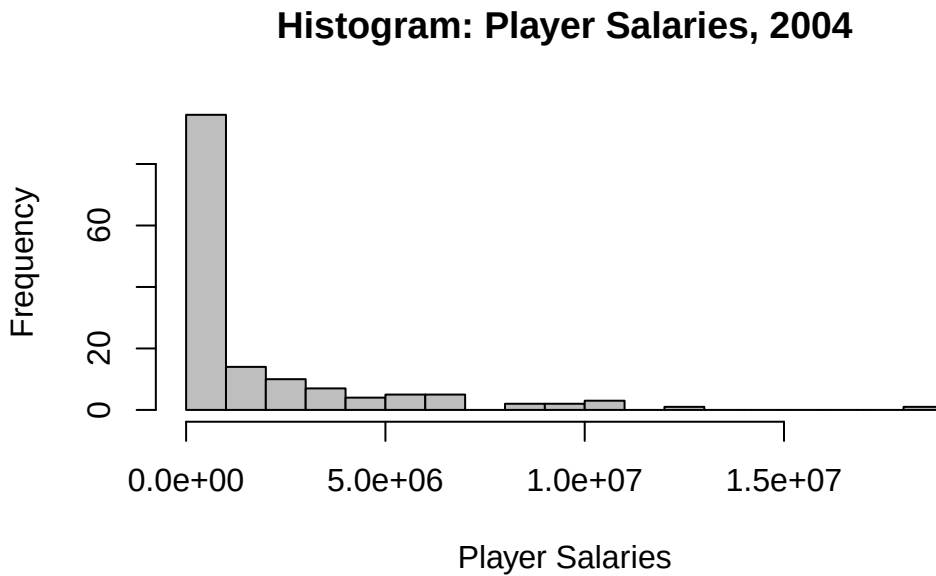
|    | dplay | pball | gbat | atbat | run | hit | secbase | thirdbase | homerun | rbi | stolenb | csteal |
|----|-------|-------|------|-------|-----|-----|---------|-----------|---------|-----|---------|--------|
| 8  | 9     | -9    | 148  | 577   | 83  | 171 | 22      | 17        | 5       | 60  | 34      | 13     |
| 10 | 1     | -9    | 5    | 0     | 0   | 0   | 0       | 0         | 0       | 0   | 0       | 0      |
| 21 | 0     | -9    | 3    | 0     | 0   | 0   | 0       | 0         | 0       | 0   | 0       | 0      |
| 31 | 61    | -9    | 154  | 564   | 56  | 148 | 31      | 7         | 4       | 49  | 3       | 3      |
| 33 | 0     | -9    | 11   | 17    | 2   | 2   | 2       | 0         | 0       | 0   | 0       | 0      |
| 50 | 19    | -9    | 23   | 90    | 20  | 21  | 4       | 0         | 9       | 23  | 0       | 0      |

|    | walk | strikeout | intwalk | hbp | sacrhit | sacrfly | gidplay |
|----|------|-----------|---------|-----|---------|---------|---------|
| 8  | 49   | 94        | 0       | 0   | 10      | 2       | 6       |
| 10 | 0    | 0         | 0       | 0   | 0       | 0       | 0       |
| 21 | 0    | 0         | 0       | 0   | 0       | 0       | 0       |
| 31 | 31   | 59        | 2       | 2   | 12      | 4       | 11      |
| 33 | 0    | 7         | 0       | 0   | 1       | 0       | 1       |
| 50 | 14   | 21        | 0       | 0   | 0       | 0       | 2       |

Let's plot the salaries from our sampled players

```
hist(baseballsrs$salary,breaks=20,col="gray",xlab="Player Salaries",  
      main="Histogram: Player Salaries, 2004")
```



## 3.2 Estimation from a SRS

Once we've selected our sample, we can use it to estimate characteristics of the population, along with their standard errors.

### 3.2.1 Estimating a population mean

As a reminder, the population mean is

A natural estimator of  $\bar{y}_U$  using a SRS of sample size  $n$  is the **sample mean**:

**Example:** In our sample of  $n = 150$  baseball players, the mean salary is

```
mean.salary<-mean(baseballsrs$salary)
mean.salary
```

```
[1] 1993886
```

Because we have access to the entire population, we can check to see how our sample mean compares to the population mean.

```
mean.pop<-mean(baseball$salary)
mean.pop
```

```
[1] 2497669
```

What happens if we take a different sample? Let's try a few. What do we need to change in R?

It turns out that with simple random sampling, the sample mean is an unbiased estimator of the population mean. That is,

As noted in Chapter 2, the finite population variance is

Under simple random sampling, the sample variance  $s^2$

is an unbiased estimator of  $\sigma^2$ . However, if we want to use  $\bar{y}$  to draw conclusions about the population, we need the variance of  $\bar{y}$ . It turns out that under simple random sampling,

(Proof: see section 2.9)

But, we generally don't know  $\sigma^2$ , so we'll use an unbiased estimator of  $\text{Var}(\bar{y})$ :

The square root of the variance of the estimator is its standard error (fun fact: standard errors are not usually unbiased).

This quantity

is called the **finite population correction factor**, or **fpc** for short.

Keep in mind that we've sampled  $\frac{n}{N}$  of the population. This is called the **sampling fraction**. Why does the sampling fraction matter?

**Example:** For the baseball SRS

We know:

With the fpc:

```
var.salary<-var(baseballsrs$salary)
var.salary
```

```
[1] 8.804137e+12
```

```
se.mean<-sqrt((var.salary/n)*(1-(n/N)))
se.mean
```

```
[1] 218283.4
```

Without the fpc:

```
se.nofpc<-sqrt(var.salary/n)
se.nofpc
```

```
[1] 242269
```

What if we had sampled fewer players? Let's try 30 instead of 150. Now:

```
set.seed(98263476)
sampsiz<-30
indexsmall<-srswor(sampsiz,N)
srssmalln<-getdata(baseball,indexsmall)
meansmall<-mean(srssmalln$salary)
meansmall
```

```
[1] 1904472
```

```
var.small<-var(srssmalln$salary)
var.small
```

```
[1] 4.76297e+12
```

```
se.small<-sqrt((var.small/sampsize)*(1-(sampsize/N)))
se.small
```

```
[1] 390883
```

```
nofpc.small<-sqrt(var.small/sampsize)
nofpc.small
```

```
[1] 398454.1
```

We may also be interested in a measure of relative variability (this will be most helpful in the future for selecting sample size). The **coefficient of variation** (CV) of  $\bar{y}$  is

The CV does not depend on the units of measurement. We can estimate the CV using

**Example:** For the baseball SRS, the estimated CV is

```
CV<-se.mean/mean.salary
CV
```

```
[1] 0.1094764
```

The salary data are quite skewed, so we should also report the median:

```
med.salary<-median(baseballsrs$salary)
med.salary
```

```
[1] 525000
```

### 3.2.2 Estimating a population total

As a reminder, the population total is

We can also express the population total as a function of the population mean

It's then natural to estimate  $t$  using

It's easy to see that

The CV doesn't change because the  $N$ s cancel!

**Example:** Let's estimate the total number of runs across the 2004 season, based on our SRS of  $n = 150$  players. First, let's start with the mean runs

```
mean.runs<-mean(baseballsrs$run)
mean.runs
```

```
[1] 24.39333
```

```
var.runs<-var(baseballsrs$run)
var.runs
```

```
[1] 998.2268
```

```
se.meanrun<-sqrt((var.runs/n)*(1-(n/N)))
se.meanrun
```

```
[1] 2.324299
```

We can now use these to find an estimate of the total number of runs, along with its standard error:

```
tot.runs<-N*mean.runs
tot.runs
```

```
[1] 19441.49
```

```
se.totrun<-N*se.meanrun
se.totrun
```

```
[1] 1852.466
```

Again, since we have the whole population we can calculate the true total number of runs:

```
pop.totrun<-N*mean(baseball$run)
pop.totrun
```

```
[1] 20515
```

### 3.2.3 Estimating a population proportion

A proportion is just a special case of a mean! Let  $p$  be the proportion of the population that has a characteristic, and define  $y_i$  as

Then,

and  $\hat{p}$  is an unbiased estimator for  $p$ .

Now let's consider the variance.

If we plug  $\hat{p}$  in for  $p$ :

**Example:** Let's estimate the population proportion of players who had at least one home run. First, we'll need to do some fiddling with the data set.

```
baseballsrs$hr<-rep(0,n)
baseballsrs$hr[baseballsrs$homerun>0]<-1
head(baseballsrs)
```

|    | ID_unit | team | leagueid | player   | salary  | pos | gplay | gstart | inning | putout | assist |
|----|---------|------|----------|----------|---------|-----|-------|--------|--------|--------|--------|
| 8  | 8       | ANA  | AL       | figgich0 | 320000  | 3B  | 148   | 80     | 2116   | 57     | 129    |
| 10 | 10      | ANA  | AL       | greggke0 | 301500  | P   | 5     | 0      | 263    | 2      | 5      |
| 21 | 21      | ANA  | AL       | rodrifr0 | 375000  | P   | 3     | 0      | 252    | 5      | 9      |
| 31 | 31      | ARI  | NL       | cintral0 | 335000  | SS  | 154   | 125    | 3297   | 141    | 383    |
| 33 | 33      | ARI  | NL       | daiglca0 | 300000  | P   | 11    | 10     | 147    | 5      | 6      |
| 50 | 50      | ARI  | NL       | sexsori0 | 8725000 | 1B  | 23    | 23     | 613    | 199    | 27     |

|    | error | dplay | pball | gbat | atbat | run | hit | secbase | thirdbase | homerun | rbi | stolenb |
|----|-------|-------|-------|------|-------|-----|-----|---------|-----------|---------|-----|---------|
| 8  | 11    | 9     | -9    | 148  | 577   | 83  | 171 | 22      | 17        | 5       | 60  | 34      |
| 10 | 0     | 1     | -9    | 5    | 0     | 0   | 0   | 0       | 0         | 0       | 0   | 0       |
| 21 | 0     | 0     | -9    | 3    | 0     | 0   | 0   | 0       | 0         | 0       | 0   | 0       |
| 31 | 15    | 61    | -9    | 154  | 564   | 56  | 148 | 31      | 7         | 4       | 49  | 3       |
| 33 | 0     | 0     | -9    | 11   | 17    | 2   | 2   | 2       | 0         | 0       | 0   | 0       |
| 50 | 1     | 19    | -9    | 23   | 90    | 20  | 21  | 4       | 0         | 9       | 23  | 0       |

|    | csteal | walk | strikeout | intwalk | hbp | sacrhit | sacrflly | gidplay | hr |
|----|--------|------|-----------|---------|-----|---------|----------|---------|----|
| 8  | 13     | 49   | 94        | 0       | 0   | 10      | 2        | 6       | 1  |
| 10 | 0      | 0    | 0         | 0       | 0   | 0       | 0        | 0       | 0  |
| 21 | 0      | 0    | 0         | 0       | 0   | 0       | 0        | 0       | 0  |
| 31 | 3      | 31   | 59        | 2       | 2   | 12      | 4        | 11      | 1  |
| 33 | 0      | 0    | 7         | 0       | 0   | 1       | 0        | 1       | 0  |
| 50 | 0      | 14   | 21        | 0       | 0   | 0       | 0        | 2       | 1  |

Now we can use our new variable to estimate the proportion of all players that had at least one homerun:

```
prop.hr<-mean(baseballsrs$hr)
prop.hr
```

```
[1] 0.5066667
```

and find its standard error

```
fpc<-1-(n/N)
fpc
```

```
[1] 0.8117942
```

```
var.prophr<-(fpc*prop.hr*(1-prop.hr))/(n-1)
var.prophr
```

```
[1] 0.001361829
```

```
se.prophr<-sqrt(var.prophr)
se.prophr
```

```
[1] 0.03690296
```

### 3.3 Sampling Weights

Earlier, we defined the **inclusion probability** ( $\pi_i$ ) for each sampling unit. This is the probability that unit  $i$  is included in the sample. In probability samples, the inclusion probabilities are used to calculate  $\bar{y}$  and  $\hat{t}$ .

To see how they are used, we need a new concept—the **sampling weight**.

 Definition: Sampling weight

The **sampling weight**

For example, with the baseball data

The sampling weight is also sometimes called the **design weight**.

For simple random samples, all sampling units have the same inclusion probability:

This means that all sampling units also have the same sampling weight:

Because all units have the same sampling weight, an SRS is a **self-weighting sample**. However, just because a sample is self-weighting does not mean it's an SRS. It's possible to have a sample design where every unit has the same inclusion probability, but not all possible samples have the same chance to be selected.

As mentioned above, we can use the sampling weights  $w_i$  to calculate  $\bar{y}$  and  $\hat{t}$  (and, thus,  $\hat{p}$ ).

For an SRS,  $w_i = N/n$ , so

We didn't need to use them in baseball example to get  $\bar{y}$ ,  $\hat{t}$ , and  $\hat{p}$ . But, survey software packages in R and SAS are written to estimate the parameters for any probability sampling design so they require the data set include sampling weights.

**Example:** Let's add the sampling weights to the baseball data.

```
baseballsrs$sampwt<-rep(N/n,n)
head(baseballsrs)
```

|   | ID_unit | team | leagueid | player   | salary | pos | gplay | gstart | inning | putout | assist |
|---|---------|------|----------|----------|--------|-----|-------|--------|--------|--------|--------|
| 8 | 8       | ANA  | AL       | figgich0 | 320000 | 3B  | 148   | 80     | 2116   | 57     | 129    |

|                                                                            |    |     |    |          |         |    |     |     |      |     |          |    |
|----------------------------------------------------------------------------|----|-----|----|----------|---------|----|-----|-----|------|-----|----------|----|
| 10                                                                         | 10 | ANA | AL | greggke0 | 301500  | P  | 5   | 0   | 263  | 2   | 5        |    |
| 21                                                                         | 21 | ANA | AL | rodrifr0 | 375000  | P  | 3   | 0   | 252  | 5   | 9        |    |
| 31                                                                         | 31 | ARI | NL | cintral0 | 335000  | SS | 154 | 125 | 3297 | 141 | 383      |    |
| 33                                                                         | 33 | ARI | NL | daiglca0 | 300000  | P  | 11  | 10  | 147  | 5   | 6        |    |
| 50                                                                         | 50 | ARI | NL | sexsori0 | 8725000 | 1B | 23  | 23  | 613  | 199 | 27       |    |
| error dplay pball gbat atbat run hit secbase thirdbase homerun rbi stolenb |    |     |    |          |         |    |     |     |      |     |          |    |
| 8                                                                          | 11 | 9   | -9 | 148      | 577     | 83 | 171 | 22  | 17   | 5   | 60       | 34 |
| 10                                                                         | 0  | 1   | -9 | 5        | 0       | 0  | 0   | 0   | 0    | 0   | 0        | 0  |
| 21                                                                         | 0  | 0   | -9 | 3        | 0       | 0  | 0   | 0   | 0    | 0   | 0        | 0  |
| 31                                                                         | 15 | 61  | -9 | 154      | 564     | 56 | 148 | 31  | 7    | 4   | 49       | 3  |
| 33                                                                         | 0  | 0   | -9 | 11       | 17      | 2  | 2   | 2   | 0    | 0   | 0        | 0  |
| 50                                                                         | 1  | 19  | -9 | 23       | 90      | 20 | 21  | 4   | 0    | 9   | 23       | 0  |
| csteal walk strikeout intwalk hbp sacrhith sacrfly gidplay hr sampwt       |    |     |    |          |         |    |     |     |      |     |          |    |
| 8                                                                          | 13 | 49  |    | 94       | 0       | 0  | 10  | 2   | 6    | 1   | 5.313333 |    |
| 10                                                                         | 0  | 0   |    | 0        | 0       | 0  | 0   | 0   | 0    | 0   | 5.313333 |    |
| 21                                                                         | 0  | 0   |    | 0        | 0       | 0  | 0   | 0   | 0    | 0   | 5.313333 |    |
| 31                                                                         | 3  | 31  |    | 59       | 2       | 2  | 12  | 4   | 11   | 1   | 5.313333 |    |
| 33                                                                         | 0  | 0   |    | 7        | 0       | 0  | 1   | 0   | 1    | 0   | 5.313333 |    |
| 50                                                                         | 0  | 14  |    | 21       | 0       | 0  | 0   | 0   | 2    | 1   | 5.313333 |    |

The sampling weight is  $N/n = 797/150 = 5.31333$ , indicating that player in our SRS represents 5.31333 players in the population of all 2004 MLB players.

We can then use the sampling weights to calculate the same estimates we did earlier, using a built in function in the **survey** package in R. We first have to define the survey design, using the **svydesign** function:

```
library(survey)
baseballdesign<-svydesign(id=~1, weights=~sampwt, fpc=rep(N,n), data=baseballsrs)
```

The **svydesign** function has several inputs:

- id
- weights
- fpc
- data

There are other inputs that we don't need right now, but we'll add them as we encounter more complicated sampling designs.

Let's look the information included in `baseballdesign`:

```
baseballdesign
```

Independent Sampling design

```
svydesign(id = ~1, weights = ~sampwt, fpc = rep(N, n), data = baseballsrs)
```

R tells us this is an Independent Sampling Design, which means there are no clusters or strata.

Now that we've set up the sampling design in R, we can use it calculate the same estimates we did before with a lot less code. We can use the functions `svymean` and `svytotal` to get  $\bar{y}$  and  $\hat{t}$ , respectively, along with their standard errors. The generic code is:

```
ybar<-svymean(~var_interest, design)
```

```
that<-svytotal(~var_interest, design)
```

**Example:** For estimating average salary and total runs in 2004 we'd use:

```
salmean<-svymean(~salary,baseballdesign)
salmean
```

|        | mean    | SE     |
|--------|---------|--------|
| salary | 1993886 | 218283 |

```
runtotal<-svytotal(~run,baseballdesign)
runtotal
```

|     | total | SE     |
|-----|-------|--------|
| run | 19442 | 1852.5 |

The same values we got before. Since  $\hat{p}$  is really a mean, we can also use `svymean` to get  $\hat{p}$  and its SE:

```
hrprop<-svymean(~hr,baseballdesign)
hrprop
```

|    | mean    | SE     |
|----|---------|--------|
| hr | 0.50667 | 0.0369 |

Again, the same values we got before.

## 3.4 Confidence Intervals

Once we select the random sample and use the resulting data to construct estimates of a population value (mean, total, proportion), the next step is typically to make some statement about the accuracy of the estimate. This is most often accomplished by constructing a **confidence interval** for the true population value.

The confidence intervals we'll construct have the same interpretation as those we constructed in classes like STAT 218, 318, 102, 212, 380.

💡 Definition: Confidence interval

A **confidence interval**

The expected proportion of intervals containing the true population value is the **coverage probability**. So, for example, an interval with a coverage probability of 0.95 means

But, there are differences. In probability sampling with a finite population, only a finite number of possible samples exist, and we know the probability with which each sample will be chosen. This means we *could* generate all possible samples, and all possible intervals. From these, we could then observe how many capture the population value, and get the *exact* coverage probability.

In real life, we can't do this

But, the confidence intervals we used in STAT 218/318/102/212/380 (which assumed an infinite population) relied on asymptotic theory—what happens as  $n \rightarrow \infty$ . For example, we know and love the Central Limit Theorem:

But we've only got a finite population! How can we use a result about  $n \rightarrow \infty$  when we've got a finite population?

Happily, there is a version of the Central Limit Theorem for SRSWOR, due to Hájek (1960). If some conditions that are *way* too technical for us to talk about are true and if

then

This means we could use this confidence interval for the population mean:

Again, we don't know  $\sigma$ ! We could substitute in  $s$  and use the standard error. And, in practice, we typically substitute  $t_{\alpha/2, df=n-1}$  for  $z_{\alpha/2}$ . In small samples, it's more conservative (wider confidence intervals) and with large samples  $t_{\alpha/2, n-1} \approx z_{\alpha/2}$  anyway. So, we'll use the interval

to calculate a  $(1 - \alpha) \times 100\%$  confidence interval for the from population mean from an SRS.

In general,

We have the same question come up that comes with the regular CLT: how large is large enough for  $n$  (and  $N$ ?) It depends

Some rules of thumb:

- If the population is approximately normal,
- 

Sugden et al. (2000) recommended the minimum sample size

Another way to check whether the distribution is too skewed to use the CLT is bootstrapping!

**Example:** Back to the baseball data. Let's get a 95% confidence interval for the mean salary of players in 2004. We saw already that based on our SRS of  $n = 150$ ,  $\bar{y} = 1993886$  and  $SE(\bar{y}) = 218283.4$ . We can use the `qt(alpha/2,df)` function to get the  $t$  multiplier in the confidence interval.

```
tcrit<-qt(0.025,n-1,lower.tail=FALSE)
lower<-mean.salary-tcrit*se.mean
upper<-mean.salary+tcrit*se.mean
lower
```

```
[1] 1562555
```

```
upper
```

```
[1] 2425217
```

Should we have even used the  $t$ -interval for the salary data? It's pretty skewed. We can find the skewness to check.

```
library(moments)
salskew<-skewness(baseballsr$salary)
salskew
```

```
[1] 2.538835
```

And using the Sudgen et al. (2000) recommendation, we need at least

```
nmin<-28 + 25*(salskew)**2
nmin
```

```
[1] 189.142
```

What about a 95% confidence interval for the total number of runs in 2004? We already found  $\hat{t} = 19441.49$  and  $SE(\hat{t}) = 1852.4661651$ .

```
lowerrun<-tot.runs-tcrit*se.totrun  
upperrun<-tot.runs+tcrit*se.totrun  
lowerrun
```

```
[1] 15780.99
```

```
upperrun
```

```
[1] 23101.98
```

Let's try a 95% confidence interval for the proportion players with at least one home run. We know  $\hat{p} = 0.5066667$  and  $SE(\hat{p}) = 0.036903$ .

```
lowerhr<-prop.hr-tcrit*se.prophr  
upperhr<-prop.hr+tcrit*se.prophr  
lowerhr
```

```
[1] 0.4337459
```

```
upperhr
```

```
[1] 0.5795874
```

It turns out the `survey` package has confidence intervals built in to the `svymean` and `svytotal` functions. Once we have created an object from either function, we can pass that object into the `confint` function. Here's the syntax:

```
confint(object,df= , level= )
```

So we could use

```
salmean<-svymean(~salary, baseballdesign)
salmean
```

```
      mean      SE
salary 1993886 218283
```

```
confint(salmean,df=n-1, level=0.95)
```

```
      2.5 %  97.5 %
salary 1562555 2425217
```

For more practice:

1. Estimating the mean number of errors committed by a player in 2004
  - draw a new sample of size  $n = 150$  (with no seed or pick your own seed, so everyone gets a different sample) from the baseball data set
  - use `svymean` to a 90% confidence interval for the mean number of errors committed by a player in 2004
2. Estimating the total number of pitchers in 2004
  - using the same sample you drew above, create a 0/1 variable that identifies whether each player in the sample is a pitcher (note that `svydesign` must be run again if a new variable is added to the data set)
  - use `svymean` to get an estimate for  $\hat{p}$
  - use `svytotal` to estimate the total number of pitchers in 2004
3. Estimate something of your choice!

## 3.5 Sample Size

Often the first question that comes up when designing a survey is how many observations are needed. This is often a pretty complicated question, and involves balancing the objectives of the study with the available budget. You also must consider that most studies have multiple characteristics being measured on each observation unit.

If the researcher can narrow down the most important variables to focus on one or two, there are four basic steps:

1. Decide how much precision you need (basically, how much variability are you willing to live with in the estimates)
2. Find an equation involving the sample size  $n$  and the precision (i.e., the margin of error)
3. Estimate anything you need to and solve for  $n$  (usually an estimate of the expected variance)
4. Go back to square one when the  $n$  turns out to be enormous.

We'll go through these steps one by one.

### 1. Decide how much precision you need

### 2. Find an equation involving $n$ and the precision

Let's start by assuming the fpc is negligible (i.e., we have an infinite population)

Now, let's assume  $\frac{n_0}{N} < 1$  (why is this a reasonable assumption?)

So, the sample size we need is

We could also consider finding the sample size to achieve a specified relative precision, call it  $r$ . The relative precision measures the distance between the estimate and the true value, scaled by the true value.

Sometimes we can calculate the required  $n$  at this stage!

**Example:** Suppose we want to estimate the proportion of 2004 baseball players that had at least one error, and we want to be able to calculate a 95% confidence interval and have a margin of error of 0.04.

### 3. Estimate anything you need to and solve for $n$

This usually involves estimating (guessing) the expected variance. Why didn't need to do this in the baseball example above?

How can we obtain the expected variance?

Another option is to use the CV for a sample of size 1:  $\sigma/\bar{y}_u$

If there are no other options, guess. An educated guess may use the suspected distribution of the data.

### 4: Go back to square one when the $n$ turns out to be enormous

**Example:** A botanical researchers wishes to design a survey to estimate the number of birch trees in a study area. The study area has been divided into 1000 plots. From previous experience, the variance in the number of trees per plot is about  $\sigma^2 \approx 45$ . What sample size should be used to estimate the total number of trees in the study area to within 500 trees of the true value with 95% confidence?

To within 1000 trees?

To within 2000 trees?

Does the finite population correction factor matter here? How do you know?

## 3.6 Systematic Sampling

Suppose we have a population with  $N$  units, and the units are numbered from 1 to  $N$  in some order. To select a sample of  $n$  units, we:

For example,

Note that

This type of sampling design is called a **systematic sample**

Why isn't this a SRS?

When will a systematic sample behave like an SRS?

When will a systematic sample not behave like an SRS?

There are a couple of advantages over the SRS:

## 3.7 Considerations for Choosing an SRS Design

### When Shouldn't You Use an SRS?

- You really should be carrying out a designed experiment
- You don't have a sampling frame or it's too expensive
- You've got additional information

### Why Should You Use an SRS?

- You don't have extra information
- Explainability is super important
- Easier computationally

## 4 Stratified Sampling

In the last chapter, we noted that an SRS isn't the best option if we have additional information that could help us come up with a more cost-effective sampling design. One such option is **stratified sampling**.

### 4.1 Overview of Stratified Sampling

#### Basic Concept: Stratification

The idea of **stratification** is

For example,

The basic set-up for stratified sampling is as follows:

- 
- 
-

What should we be thinking about to decide whether or not to stratify?

- Is there a chance we could draw a really bad sample?
- Do we want to draw conclusions about a particular subgroup of the population (especially with a particular precision)?
- Is it more convenient or cheaper?
- Can we construct the strata in a way that our estimates will be more precise?

If the answer to any of the above is yes, stratification is probably a good idea.

**Example:** In the last section, we selected a SRS of  $n = 150$  baseball players from the 2004 MLB season. One of the variables we considered was salary. We saw that there was a lot of variability in the data, possibly due to the strong right skew. Is there also variability due to other factors? Maybe there is variability due to league as well.

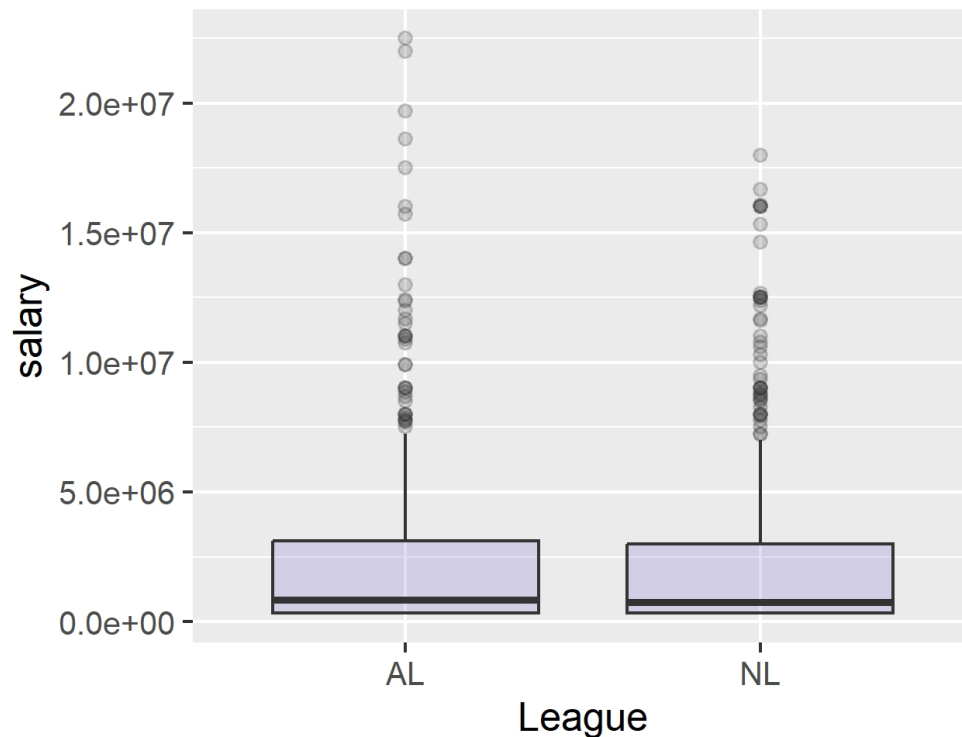


Figure 4.1: Baseball Salary by League

It looks like there is more variability among salary for the American League. Let's use League as our stratification variable. It meets the requirement:

We'll stick with a sample of  $n = 150$  players. In the population,  $372/797 = 0.46675$  players are in the AL and  $425/797 = 0.53325$  players are in the NL. So, we'll sample 70 AL players and 80 NL players.

Another way to think about this ...

When we found mean salary before, the SE was 218283.4. When I drew a stratified sample with the allocation above, the SE is 200945. So perhaps stratifying by league could be helpful in increasing precision.

## 4.2 How Does Stratified Sampling Work?

We're going to start by dividing the population ( $N$  sampling units) into  $H$  mutually exclusive and exhaustive strata,

- 
- 
- 

Then, we select an independent probability sample from each stratum.

Some choices have to be made here, namely how to allocate the sample across the strata. In the baseball example, we used **proportional allocation**:

There are other methods of allocation as well, and we'll talk about these later.

What's important is that

We could use any sampling design (SRS, Systematic, etc.) within a stratum, and they don't all need to be the same. For simplicity, we'll assume a SRSWOR is selected within each stratum.

With this assumption of SRSWOR, we are carrying out **stratified random sampling**, and the biggest adjustment from SRS to stratified is really bookkeeping—there are a lot more subscripts to keep track of.

As a reminder, we assume

We'll start with new notation for the population quantities:

- $N_h$

- $y_{hj}$

- $t_h$

- $t$

- $\bar{y}_{h\mathcal{U}}$

- $\bar{y}_{\mathcal{U}}$

- $S_h^2 = \sigma_h^2$

The sample quantities will also have subscripts:

- $\mathcal{S}_h$

- $n_h$

- $n$

- $\bar{y}_h$

- $\hat{t}_h$

- $s_h^2$

## 4.3 Estimation with Stratified Random Samples

Suppose we sampled only the  $h^{th}$  stratum:

This leads to an estimate for entire population total

and entire population mean

The estimated population mean is really just a weighted average of the sample stratum means, because

Because we took a SRS in each stratum, all the properties of SRS estimates are retained:

- **Unbiasedness**

- Variance of the estimators

- Confidence intervals

**Example:** In Homework 1, you worked with some data on the number of publications for a SRS of  $n = 50$  faculty, out of a total of  $N = 807$  faculty members. We'll use the areas biological sciences, physical sciences, social sciences, and humanities as the strata.

| Stratum             | Num of Faculty in Stratum | Num of Faculty in Sample |
|---------------------|---------------------------|--------------------------|
| Biological sciences | 102                       | 7                        |
| Physical sciences   | 310                       | 19                       |
| Social sciences     | 217                       | 13                       |
| Humanities          | 178                       | 11                       |
| Total               | 807                       | 50                       |

The actual publication data for each stratum is in the R code snippets below.

Let's set up a table similar to one shown in Example 3.3

| Stratum    | $N_h$ | $n_h$ | $\bar{y}_h$ | $s_h^2$ | $\hat{t}_h = N_h \bar{y}_h$ | $(1 - \frac{n_h}{N_h}) N_h^2 \frac{s_h^2}{n_h}$ |
|------------|-------|-------|-------------|---------|-----------------------------|-------------------------------------------------|
| Bio sci    | 102   | 7     | 3.14        | 6.81    |                             |                                                 |
| Phy sci    | 310   | 19    | 2.105       | 8.21    |                             |                                                 |
| Soc sci    | 217   | 13    | 1.231       | 4.359   |                             |                                                 |
| Humanities | 178   | 11    | 0.4545      | 0.8727  |                             |                                                 |
| Total      | 807   | 50    |             |         |                             |                                                 |

So, our estimated total number of publications is

with standard error

So, a 95% confidence interval for the total number of publications is

In Chapter 3 of the notes, we observed that a proportion is just a mean where the only values the variable can take on are 0 and 1. We'll use the same idea here and define

Then we can estimate the overall proportion of the population with the characteristic as

with estimated variance

If we want to estimate the total number of those in the population with the characteristic we can use

**Example:** Let's go back to the publications data and consider the proportion of faculty with at least one publication.

| Stratum    | $N_h$ | $n_h$ | $N_h/N$ | $\bar{y}_h = \hat{p}_h$ | $(1 - \frac{n_h}{N_h})(\frac{N_h}{N})^2(\frac{\hat{p}_h(1-\hat{p}_h)}{n_h-1})$ |
|------------|-------|-------|---------|-------------------------|--------------------------------------------------------------------------------|
| Bio sci    | 102   | 7     | 0.1264  | 0.857                   |                                                                                |
| Phy sci    | 310   | 19    | 0.3841  | 0.474                   |                                                                                |
| Soc sci    | 217   | 13    | 0.2689  | 0.308                   |                                                                                |
| Humanities | 178   | 11    | 0.2206  | 0.273                   |                                                                                |
| Total      | 807   | 50    |         |                         |                                                                                |

So,

$$\hat{p}_{str} = \hat{p}_{\{str\}}$$

and

$$SE(\hat{p}_{str}) =$$

We can also estimate the total number of faculty with at least one publication and its standard error.

## 4.4 Sampling Weights

With simple random samples, we observed that the sampling weights were

because all sampling units have the same inclusion probability

and the sampling weights are

With stratified sampling, the inclusion probability may not be the same for all units (they will be the same for all units within the same stratum).

The **sampling weight for unit  $j$**  of stratum  $h$  is

The interpretation of the sampling weight is the same as it was with SRS!

Just as before, the sampling weight is the inverse of the inclusion probability:

We can again re-write the population estimates using the sampling weights

In the last section, we observed that simple random samples are **self-weighting** samples, because each sample unit has the same weight. Stratified random samples will be self-weighting samples if

Even if this is true, however, we still need to acknowledge the stratification in the estimation because it will make a difference in the variances.

We can use the **survey** package to calculate estimates using the **svymean** and **svytotal** functions, but we do need to specify the stratification variable and the sampling weights in the sampling design (i.e., in **svydesign**). Let's look more carefully at the R code for the publication data. We also have a new function, **svyby**.

First, we need to get the data in R. I am **not** an elegant R coder, I'm sure there are more efficient ways to do this.

```
biopubs<-c(rep(0,1),rep(1,2),rep(3,1),rep(5,2),rep(7,1))
biopubs
```

```
[1] 0 1 1 3 5 5 7
```

```
phypubs<-c(rep(0,10),rep(1,2),rep(3,1),rep(4,2),rep(5,1),rep(6,1),rep(8,2))
socpubs<-c(rep(0,9),rep(2,1),rep(4,2),rep(6,1))
humpubs<-c(rep(0,8),rep(1,2),rep(3,1))
```

```
biopubsdf<-data.frame(biopubs)
phypubsdf<-data.frame(phypubs)
socpubsdf<-data.frame(socpubs)
humpubsdf<-data.frame(humpubs)
```

```
biopubsdf$stratum<-"Biological"
biopubsdf$pubs<-biopubsdf$biopubs
biopubsdf$Nh<-102
biopubsdf$samp<-7
biopubsdf<-biopubsdf[,c(2,3,4,5)]
```

```
biopubsdf
```

|   | stratum    | pubs | Nh  | samp |
|---|------------|------|-----|------|
| 1 | Biological | 0    | 102 | 7    |
| 2 | Biological | 1    | 102 | 7    |
| 3 | Biological | 1    | 102 | 7    |
| 4 | Biological | 3    | 102 | 7    |
| 5 | Biological | 5    | 102 | 7    |
| 6 | Biological | 5    | 102 | 7    |
| 7 | Biological | 7    | 102 | 7    |

```
phypubsdf$stratum<-"Physical"
phypubsdf$pubs<-phypubsdf$phypubs
phypubsdf$Nh<-310
phypubsdf$samp<-19
phypubsdf<-phypubsdf[,c(2,3,4,5)]
socpubsdf$stratum<-"Social"
socpubsdf$pubs<-socpubsdf$socpubs
socpubsdf$Nh<-217
```

```

socpubsdf$samp<-13
socpubsdf<-socpubsdf[,c(2,3,4,5)]
humpubsdf$stratum<-"Humanities"
humpubsdf$pubs<-humpubsdf$humpubs
humpubsdf$Nh<-178
humpubsdf$samp<-11
humpubsdf<-humpubsdf[,c(2,3,4,5)]

pubsdf<-rbind(biopubsdf,phypubsdf,socpubsdf,humpubsdf)
pubsdf

```

|    | stratum    | pubs | Nh  | samp |
|----|------------|------|-----|------|
| 1  | Biological | 0    | 102 | 7    |
| 2  | Biological | 1    | 102 | 7    |
| 3  | Biological | 1    | 102 | 7    |
| 4  | Biological | 3    | 102 | 7    |
| 5  | Biological | 5    | 102 | 7    |
| 6  | Biological | 5    | 102 | 7    |
| 7  | Biological | 7    | 102 | 7    |
| 8  | Physical   | 0    | 310 | 19   |
| 9  | Physical   | 0    | 310 | 19   |
| 10 | Physical   | 0    | 310 | 19   |
| 11 | Physical   | 0    | 310 | 19   |
| 12 | Physical   | 0    | 310 | 19   |
| 13 | Physical   | 0    | 310 | 19   |
| 14 | Physical   | 0    | 310 | 19   |
| 15 | Physical   | 0    | 310 | 19   |
| 16 | Physical   | 0    | 310 | 19   |
| 17 | Physical   | 0    | 310 | 19   |
| 18 | Physical   | 1    | 310 | 19   |
| 19 | Physical   | 1    | 310 | 19   |
| 20 | Physical   | 3    | 310 | 19   |
| 21 | Physical   | 4    | 310 | 19   |
| 22 | Physical   | 4    | 310 | 19   |
| 23 | Physical   | 5    | 310 | 19   |
| 24 | Physical   | 6    | 310 | 19   |
| 25 | Physical   | 8    | 310 | 19   |
| 26 | Physical   | 8    | 310 | 19   |
| 27 | Social     | 0    | 217 | 13   |
| 28 | Social     | 0    | 217 | 13   |
| 29 | Social     | 0    | 217 | 13   |
| 30 | Social     | 0    | 217 | 13   |
| 31 | Social     | 0    | 217 | 13   |

|    |            |   |     |    |
|----|------------|---|-----|----|
| 32 | Social     | 0 | 217 | 13 |
| 33 | Social     | 0 | 217 | 13 |
| 34 | Social     | 0 | 217 | 13 |
| 35 | Social     | 0 | 217 | 13 |
| 36 | Social     | 2 | 217 | 13 |
| 37 | Social     | 4 | 217 | 13 |
| 38 | Social     | 4 | 217 | 13 |
| 39 | Social     | 6 | 217 | 13 |
| 40 | Humanities | 0 | 178 | 11 |
| 41 | Humanities | 0 | 178 | 11 |
| 42 | Humanities | 0 | 178 | 11 |
| 43 | Humanities | 0 | 178 | 11 |
| 44 | Humanities | 0 | 178 | 11 |
| 45 | Humanities | 0 | 178 | 11 |
| 46 | Humanities | 0 | 178 | 11 |
| 47 | Humanities | 0 | 178 | 11 |
| 48 | Humanities | 1 | 178 | 11 |
| 49 | Humanities | 1 | 178 | 11 |
| 50 | Humanities | 3 | 178 | 11 |

```
pubsdf$sampwt<-with(pubsdf,Nh/samp)
pubsdf
```

|    | stratum    | pubs | Nh  | samp | sampwt   |
|----|------------|------|-----|------|----------|
| 1  | Biological | 0    | 102 | 7    | 14.57143 |
| 2  | Biological | 1    | 102 | 7    | 14.57143 |
| 3  | Biological | 1    | 102 | 7    | 14.57143 |
| 4  | Biological | 3    | 102 | 7    | 14.57143 |
| 5  | Biological | 5    | 102 | 7    | 14.57143 |
| 6  | Biological | 5    | 102 | 7    | 14.57143 |
| 7  | Biological | 7    | 102 | 7    | 14.57143 |
| 8  | Physical   | 0    | 310 | 19   | 16.31579 |
| 9  | Physical   | 0    | 310 | 19   | 16.31579 |
| 10 | Physical   | 0    | 310 | 19   | 16.31579 |
| 11 | Physical   | 0    | 310 | 19   | 16.31579 |
| 12 | Physical   | 0    | 310 | 19   | 16.31579 |
| 13 | Physical   | 0    | 310 | 19   | 16.31579 |
| 14 | Physical   | 0    | 310 | 19   | 16.31579 |
| 15 | Physical   | 0    | 310 | 19   | 16.31579 |
| 16 | Physical   | 0    | 310 | 19   | 16.31579 |
| 17 | Physical   | 0    | 310 | 19   | 16.31579 |
| 18 | Physical   | 1    | 310 | 19   | 16.31579 |
| 19 | Physical   | 1    | 310 | 19   | 16.31579 |

|    |            |   |     |    |          |
|----|------------|---|-----|----|----------|
| 20 | Physical   | 3 | 310 | 19 | 16.31579 |
| 21 | Physical   | 4 | 310 | 19 | 16.31579 |
| 22 | Physical   | 4 | 310 | 19 | 16.31579 |
| 23 | Physical   | 5 | 310 | 19 | 16.31579 |
| 24 | Physical   | 6 | 310 | 19 | 16.31579 |
| 25 | Physical   | 8 | 310 | 19 | 16.31579 |
| 26 | Physical   | 8 | 310 | 19 | 16.31579 |
| 27 | Social     | 0 | 217 | 13 | 16.69231 |
| 28 | Social     | 0 | 217 | 13 | 16.69231 |
| 29 | Social     | 0 | 217 | 13 | 16.69231 |
| 30 | Social     | 0 | 217 | 13 | 16.69231 |
| 31 | Social     | 0 | 217 | 13 | 16.69231 |
| 32 | Social     | 0 | 217 | 13 | 16.69231 |
| 33 | Social     | 0 | 217 | 13 | 16.69231 |
| 34 | Social     | 0 | 217 | 13 | 16.69231 |
| 35 | Social     | 0 | 217 | 13 | 16.69231 |
| 36 | Social     | 2 | 217 | 13 | 16.69231 |
| 37 | Social     | 4 | 217 | 13 | 16.69231 |
| 38 | Social     | 4 | 217 | 13 | 16.69231 |
| 39 | Social     | 6 | 217 | 13 | 16.69231 |
| 40 | Humanities | 0 | 178 | 11 | 16.18182 |
| 41 | Humanities | 0 | 178 | 11 | 16.18182 |
| 42 | Humanities | 0 | 178 | 11 | 16.18182 |
| 43 | Humanities | 0 | 178 | 11 | 16.18182 |
| 44 | Humanities | 0 | 178 | 11 | 16.18182 |
| 45 | Humanities | 0 | 178 | 11 | 16.18182 |
| 46 | Humanities | 0 | 178 | 11 | 16.18182 |
| 47 | Humanities | 0 | 178 | 11 | 16.18182 |
| 48 | Humanities | 1 | 178 | 11 | 16.18182 |
| 49 | Humanities | 1 | 178 | 11 | 16.18182 |
| 50 | Humanities | 3 | 178 | 11 | 16.18182 |

With data in place, we can now find some estimates for the whole population of 807 faculty members:

```
library(survey)

pubsdes<-svydesign(id=~1,strata=~stratum,weights=~sampwt,fpc=~Nh,data=pubsdf)
strmean<-tapply(pubsdf$pubs,pubsdf$stratum,mean)
strmean
```

| Biological | Humanities | Physical  | Social    |
|------------|------------|-----------|-----------|
| 3.1428571  | 0.4545455  | 2.1052632 | 1.2307692 |

```
strvar<-tapply(pubsdf$pubs,pubsdf$stratum,var)
strvar
```

```
Biological Humanities Physical Social
6.8095238 0.8727273 8.2105263 4.3589744
```

```
pubsstrat<-svyby(~pubs,by=~stratum,pubsdes,svymean, keep.var=TRUE)
confint(pubsstrat,df=50-4)
```

```

          2.5 %    97.5 %
Biological 1.22687120 5.058843
Humanities -0.09463164 1.003723
Physical   0.82324081 3.387286
Social     0.10064302 2.360895
```

```
svytotal(~pubs,pubsdes)
```

```

total SE
pubs 1321.2 256.15
```

```
pubsdf$yes<-rep(0,50)
pubsdf$yes[pubsdf$pubs>0] <- 1
pubsdes<-svydesign(id=~1,strata=~stratum,weights=~sampwt,fpc=~Nh,data=pubsdf)
svymean(~yes,pubsdes)
```

```

mean SE
yes 0.43319 0.0658
```

```
svytotal(~yes,pubsdes)
```

```

total SE
yes 349.59 53.128
```

## 4.5 Sample Size and Allocation

With simple random samples, we only had one design feature we needed to determine: the sample size  $n$ . With stratified sampling, we need to determine:

It's actually easier to look at these in reverse order. We'll start by assuming the strata have already been defined, and consider how to allocate observations among them.

### 4.5.1 Allocation in Stratified Random Sampling

The general set-up is

There are several options for determining how many observations to sample in each stratum. Some allocation strategies deal with the whole population, others deal with subpopulations. Yet others are a compromise between the two. The best allocation strategy depends on the goals of the study.

#### 4.5.1.1 Whole Population Allocation

With whole population allocation, we first determine how many sampling units we can collect/afford in the entire sample. That is, we first determine  $n$ . Then, we turn attention to how to divide  $n$  across the strata. Whole population allocation is best when the objective of the study is to estimate a characteristic for the entire population. There are several factors that affect allocation rules

The simplest whole population method is **proportional allocation**.

##### Proportion Allocation

In this situation, the inclusion probability for unit  $j$  in stratum  $h$  is

and is the same across all strata. This also implies

One note: rounding in proportional allocation can lead to unequal (but approximately equal) sampling weights.

**Example:** With the baseball data, let's consider stratifying by position:

| Stratum $h$ | Stratum Total $N_h$ | Stratum Sample Size $n_h$ | Sample Weight $w_{hj}$ |
|-------------|---------------------|---------------------------|------------------------|
| 1 (1B)      | 66                  |                           |                        |
| 2 (2B)      | 50                  |                           |                        |
| 3 (3B)      | 53                  |                           |                        |
| 4 (C)       | 61                  |                           |                        |
| 5 (CF)      | 43                  |                           |                        |
| 6 (LF)      | 49                  |                           |                        |
| 7 (P)       | 376                 |                           |                        |
| 8 (RF)      | 54                  |                           |                        |
| 9 (SS)      | 45                  |                           |                        |
| Total       | 797                 | 150                       |                        |

When the strata are large enough,

Proportional allocation tends to provide optimal estimates (optimal in the sense of minimum variance) when

Proportional allocation is a good choice if

If the variances are not approximately equal across strata, then **Neyman allocation** is an option.

#### 💡 Neyman Allocation

Suppose the within-stratum variances for  $y_{hj}$  ( $\sigma_h^2$ ) vary across strata. To obtain a similar level of precision in each stratum, we'd need a higher sample size in strata that are more variable. For example, consider the estimate of stratum total,  $\hat{t}_h$ . We saw earlier that

Neyman allocation suggests allocating stratum sample size in proportion to

It turns out (proof not shown) that the variance of an estimate for population mean or total (for a fixed total  $n$ ) is minimized if

and we can see the logic of this by equating

**Tiny Example:** A population consists of three strata of sizes  $N_1 = 150$ ,  $N_2 = 90$ , and  $N_3 = 120$ , so  $N = 360$ . Based on previous studies, we expect the standard deviations within each strata to be  $\sigma_1 \approx 100$ ,  $\sigma_2 \approx 200$ , and  $\sigma_3 \approx 300$ . We know we can sample a total of  $n = 12$  sampling units.

Proportional allocation:

Neyman allocation:

Neyman allocation tends to provide optimal (in the sense of minimum variance) estimates when

Considerations for selecting Neyman allocation:

Neyman allocation is a special case of **optimal allocation**, which considers both variability and cost, and is often used when the sampling units themselves are of different sizes. Larger units are likely to be more variable, and are sampled at a higher rate.

**Examples:** Sampling units that vary in size could be

To build in variable cost, suppose data collection costs  $c_h$  vary across strata.

#### 💡 Optimal allocation

In **optimal allocation**, the sample size is allocated to strata in proportion to

So again, we're going to equate

and get the allocation rule

With optimal allocation we see sample size for stratum  $h$  ( $n_h$ ) is larger in strata where:

**Example:** We're designing a study where the objective is to estimate the fraction of UNL students on academic probation that have been attending academic coaching sessions. Here's our sample design

- **Frame:**
- **Stratum 1:** Students meeting with their advisors (based on notes in Student Success Hub)
  - $N_1 = 750, p_1 = 0.90, c_1 = \$100$
- **Stratum 2:** Students not meeting with their advisors (based on no notes in Student Success Hub)
  - $N_2 = 100, p_2 = 0.35, c_2 = \$400$
- **Sample size:**  $n = 100$

Here's how optimal allocation would work:

| Stratum $h$ | $N_h$ | $\sigma_h^2 \approx$<br>$p_h(1 - p_h)$ | $c_h$ | $N_h \sigma_h / \sqrt{c_h}$ | $\frac{N_h \sigma_h / \sqrt{c_h}}{\sum_{h=1}^H N_h \sigma_h / \sqrt{c_h}}$ | $n_h$     |
|-------------|-------|----------------------------------------|-------|-----------------------------|----------------------------------------------------------------------------|-----------|
| 1: Notes    | 750   |                                        | 100   |                             |                                                                            |           |
| 2: No Notes | 100   |                                        | 400   |                             |                                                                            |           |
| Total       | 850   |                                        |       |                             |                                                                            | $n = 100$ |

To see a summary of whole population allocation strategies, let's again think about equating

and look at what terms cancel out in particular circumstances. Then we can make the following summary

#### 4.5.1.2 Subpopulation Allocation

With subpopulation allocation, our main interest is in estimating stratum characteristics

One warning: **subpopulation allocation generally leads to less precise whole population estimates.**

If stratum costs and variances are equal, then **equal allocation** is an option. With equal allocation we assume

##### Equal allocation

The equal allocation rule is to divide  $n$  equally across the  $H$  strata. So,

**Example:**

### 💡 Allocation for specific precision

If our real focus is on estimating one (or two) specific subpopulation values and  $\sigma_h^2$  varies across strata, we can

Then, once each stratum sample size is determined, sum to get the total sample size:

This made lead to sample sizes that exceed the allowable budget. Some options for adjusting sample sizes when initial calculations look too high:

- Relax specifications for desired precision and recalculate stratum sample sizes. This can involve widening the allowable margin of error or decreasing the confidence level
- Decide the sample size you can afford and reduce stratum sample sizes proportionally (recheck the margin of error to make sure the confidence interval would still be useful)
- It's probably going to take some fiddling

#### 4.5.1.3 Compromise Allocation

Suppose the objective of the study is to get reasonable estimates of both whole and subpopulation values. There are allocation rules that fall somewhere between optimal for stratum estimates and optimal for population estimates. It's easiest to visualize with a graph.

Assuming equal variances and cost

The compromise is **square root allocation**.

### 💡 Square root allocation

It's a compromise in that

- More sampling units are allocated to small strata than would be with proportional allocation
- More sampling units are allocated to large strata than would be with equal allocation

**Example:** Let's compare the allocations among the baseball positions (and use  $n = 153$ , just so equal allocation gives us an integer)

| Stratum $h$ | Stratum Total<br>$N_h$ | $\sqrt{N_h}$ | Proportional<br>$n_h$ | Equal $n_h$ | Square root $n_h$ |
|-------------|------------------------|--------------|-----------------------|-------------|-------------------|
| 1 (1B)      | 66                     | 8.124        |                       | 17          |                   |
| 2 (2B)      | 50                     | 7.071        |                       | 17          |                   |
| 3 (3B)      | 53                     | 7.280        |                       | 17          |                   |
| 4 (C)       | 61                     | 7.810        |                       | 17          |                   |
| 5 (CF)      | 43                     | 6.557        |                       | 17          |                   |
| 6 (LF)      | 49                     | 7            |                       | 17          |                   |
| 7 (P)       | 376                    | 19.391       |                       | 17          |                   |
| 8 (RF)      | 54                     | 7.348        |                       | 17          |                   |
| 9 (SS)      | 45                     | 6.708        |                       | 17          |                   |
| Total       | 797                    | 77.289       |                       | 153         |                   |

There are other strategies as well! All of these are just guidelines, and you can mix features of different allocation rules.

**Example:** Say you have 5 strata and can afford a total of  $n = 4000$  observations. You may allocate 400 to each strata to achieve a margin of error of 5% percentage points on strata confidence intervals for proportions. You've only allocated  $5 \times 400 = 2000$  observations, so you can 'spend' the remaining 2000 proportionally or optimally.

There are methods for determining total sample size as well, and those are derived similarly to what we did with simple random samples. They are complicated by having multiple variances for the multiple strata. More details may be found on page 96.

## 4.5.2 Defining Strata

We started our discussion of selecting sampling design elements by stating that it's easier to work in reverse order, and imagine that we already have defined strata. Now we turn our attention to actually selecting the strata.

First, an important note: stratification almost never hurts, even if we stratify poorly.

However, stratification will be most efficient when

This means

In general, more information = more strata.

**Example:** (3.13) The point in time count of people experiencing homelessness is carried out on a single night on one of the last 10 nights in January. Large cities mean large areas to visit. Here's how NYC does it:

### 4.5.3 Allocation Examples

For all examples, we'll be keeping in mind the general allocation rule

$$n_h = \frac{n N_h s_h / \sqrt{c_h}}{\sum_{i=1}^H N_h s_h / \sqrt{c_h}}$$

6/24. Suppose that a city has 90,000 dwelling units, of which 35,000 are houses, 45,000 are apartments, and 10,000 are condominiums.

- (a) You believe that the mean electricity usage is about twice as much for houses as for apartments or condominiums, and that the standard deviation is proportional to the mean so that  $S_1 = 2S_2 = 2S_3$ . How would you allocate a stratified sample of 900 observations if you wanted to estimate the mean electricity consumption for all households in the city?
- (b) Now suppose that you take a stratified random sample with proportional allocation and want to estimate the overall proportion of households in which energy conservation is practiced. If 45% of house dwellers, 25% of apartment dwellers, and 3% of condominium residents practice energy conservation, what is  $p$  for the population? What gain would the stratified sample with proportional allocation offer over an SRS? That is, what is  $V_{\text{prop}}(\hat{p}_{\text{str}})/V_{\text{SRS}}(\hat{p}_{\text{SRS}})$ ?

8/32. A public opinion researcher has a budget of \$150,000 for interviewing households in a survey. She knows that 90% of all households have telephones. Telephone interviews cost \$100 each; in-person interviews cost \$300 each if all interviews are conducted in person, and \$400 each if only nonphone households are interviewed in person (because there will be extra travel costs). Assume that the variances in the phone and nonphone groups are similar. How many households should be interviewed in each group if

- (a) All households are interviewed in person.
- (b) Households with a phone are contacted by telephone and households without a phone are contacted in person.

17/36. Lydersen and Ryg (1991) used stratification techniques to estimate ringed seal populations in Svalbard fjords. The 200 km<sup>2</sup> study area was divided into three zones: Zone 1, outer Sassenfjorden, was covered with relatively new ice during the study period in March, 1990, and had little snow cover; Zone 3, Tempelfjorden, had a stable ice cover throughout the year; Zone 2, inner Sassenfjorden, was intermediate between the stable Zone 3 and unstable Zone 1. Ringed seals need good ice to establish territories with breathing holes, and snow cover enables females to dig out birth lairs. Thus, it was thought that the three zones would have different seal densities. The investigators took a stratified random sample of 20% of the 200 1-km<sup>2</sup>. The following table gives the number of plots, and the number of plots sampled, in each zone, along with the sample mean (and variance) number of breathing holes found in each zone:

| Zone  | Number of Plots | Plots Sampled | $\bar{y}_h$ | $s_h^2$ |
|-------|-----------------|---------------|-------------|---------|
| 1     | 68              | 17            | 1.765       | 3.316   |
| 2     | 84              | 12            | 4.417       | 11.538  |
| 3     | 48              | 11            | 10.545      | 46.073  |
| Total | 200             | 40            |             |         |

- Estimate the total number of breathing holes in the study region, along with its standard error.
- If you were designing this survey, how would you allocate observations to strata if the goal is to estimate the total number of breathing holes? If the goal is to compare the density of breathing holes in the three zones (want equal precision for  $\bar{y}_h$  in the three zones)?
- The data are in the file `seals.csv` in Canvas. Use R to verify your answer in (a).

27/70. Suppose that, instead of emailing all 15,769 persons in the current membership list of the ASA as described in Exercise 13 of Chapter 1, the researchers wanted to take a stratified random sample of 1,100 of the 5,803 female and 9,966 male ASA members. What should the sample sizes for men and women be if:

- (a) Proportional allocation is used?
- (b) Neyman allocation is used, and it is assumed that  $S^2_{\text{women}} = (11/5) \times S^2_{\text{men}}$ ?

33/33. A stratified sample is being designed to estimate the prevalence  $p$  of a rare characteristic, say the proportion of residents of Milwaukee, Wisconsin, who have Lyme disease. Stratum 1, with  $N_1$  units, has a high prevalence of the characteristic; stratum 2, with  $N_2$  units, has low prevalence. Assume that the cost to sample a unit (for example, the cost to select a person for the sample and determine they have Lyme disease) is the same for each stratum, and that at most 2000 persons are to be sampled.

- (a) Let  $p_1$  and  $p_2$  be the proportions in stratum 1 and stratum 2 with the rare characteristic. If  $p_1 = 0.10, p_2 = 0.03$ , and  $N_1/N = 0.4$ , what are  $n_1$  and  $n_2$  under optimal allocation?
- (b) If  $p_1 = 0.10, p_2 = 0.03$ , and  $N_1/N = 0.4$ , what is  $V(\hat{p}_{\text{str}})$  under proportional allocation? Under optimal allocation? What is the variance if you take an SRS of 2000 units from the population?